HOLY ANGEL UNIVERSITY

THESIS ADVISER CONSULTATION SCHEDULER

A Capstone Project
Presented to the Faculty of the
School of Computing
Holy Angel University



In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Information Technology with Area of Specialization in Web Development

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Final manuscript — complete, for the final oral defense.
Dummy document generated for testing the HAU-SOC Thesis & Capstone prototype. All names, data and references are placeholders.

SCHOOL OF COMPUTING

APPROVAL SHEET

This capstone entitled **"THESIS ADVISER CONSULTATION SCHEDULER"**, prepared and submitted in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology with Area of Specialization in Web Development has been examined and is recommended for acceptance and oral examination.

Prof. Alpha
Adviser

ORAL EXAMINATION

Approved by the committee of oral examiners on
[Date of defense]

Prof. Foxtrot
Panel Chair

Prof. Charlie
Panel Member

[Name of Panel Member]
Panel Member

APPROVAL

Accepted and approved in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology with Area of Specialization in Web Development.

Prof. Alpha
Program Chair/Coordinator,
Information Technology

Dean Delta
Dean, School of Computing

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Thesis Adviser Consultation Scheduler

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Abstract

This study aimed to develop and evaluate a web-based consultation scheduler for capstone groups and their advisers. Consultation requests arrive through several channels, meetings are double-booked or forgotten, and no shared record of consultations exists. The scheduler replaced message-based scheduling for the groups who used it, prevented double bookings, and was rated above average in usability (SUS 81.5). Two groups booked the same slot 200 times in parallel; every attempt produced exactly one booking.

Keywords: capstone, web application, ISO/IEC 25010, usability

THESIS ADVISER CONSULTATION SCHEDULER

Introduction

Capstone groups meet their advisers weekly, yet most consultations are arranged through group chats and personal messages. Advisers who handle several groups receive overlapping requests, meetings are missed when a message is buried, and neither side keeps a reliable record of what was agreed.

A scheduler that lets the adviser publish available slots and lets each group book one would remove the back-and-forth, remind both sides before the meeting, and keep a short record of every consultation that the group can refer to when it revises its work.

The problem this study addresses is the following: consultation requests arrive through several channels, meetings are double-booked or forgotten, and no shared record of consultations exists. This capstone project therefore proposes the Consultation Scheduler.

Review of Related Literature

This section reviews studies and existing systems related to the Consultation Scheduler and the methods used to evaluate it.

Appointment Scheduling in Higher Education

Studies of online appointment systems for advising offices report fewer missed meetings once students choose from published slots instead of requesting times by message (Placeholder & Author, 2023). The main design lesson is that availability must be owned by the adviser and visible to every group at once.

Reminder Timing and Attendance

Reminders sent a day and an hour before an appointment reduce no-shows more than a single reminder (Sample, 2022). Email is adequate for a university audience that already reads institutional mail on their phones.

Records of Academic Advising

Short structured notes after each meeting help students act on advice and help advisers see progress across weeks (Example & Writer, 2024).

Usability Evaluation of Scheduling Tools

The System Usability Scale is widely used to evaluate booking tools because the tasks are short and repeated (Placeholder, 2020).

Synthesis

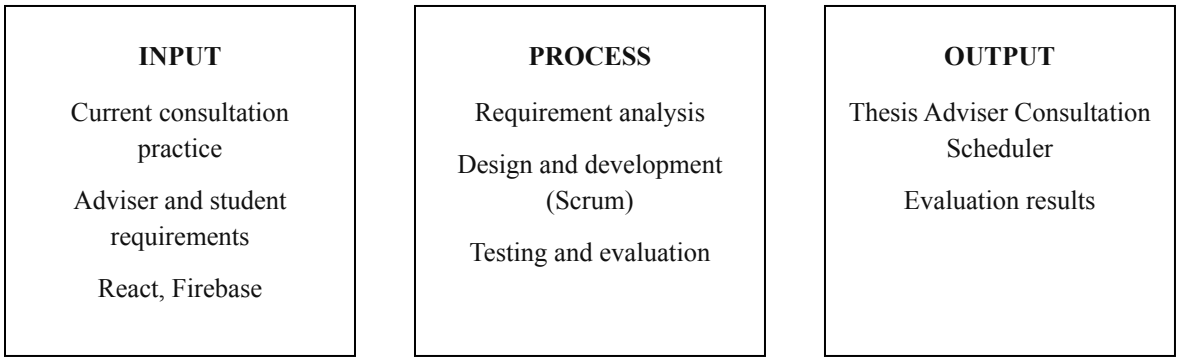
The reviewed studies show that the problem is common and that a focused, web-based solution can address it at low cost. None of the systems reviewed fits the setting of this study exactly, which is the gap the Consultation Scheduler fills.

Conceptual Framework

The study follows the Input–Process–Output model shown in Figure 1.

Figure 1

Conceptual Framework of the Study



Objectives of the Study

The general objective of the study is to develop and evaluate a web-based consultation scheduler for capstone groups and their advisers. Specifically, the study aims to:

1. analyze how capstone consultations are currently requested, scheduled and recorded, and identify the problems;
2. design and develop a system in which advisers publish consultation slots and groups book, cancel and reschedule them;
3. send email reminders to the group and the adviser one day and one hour before each consultation;
4. keep a consultation record with the topics discussed and the agreed next steps, exportable per group; and
5. evaluate the system using ISO/IEC 25010 and the System Usability Scale.

Scope and Delimitations

The system covers consultations between capstone groups and their advisers in the School of Computing. Advisers publish weekly availability; a group may hold one upcoming booking per adviser at a time. Reminders are sent by email only.

The study does not cover grading, class scheduling or video conferencing; meetings take place face to face or on the university's existing video platform, whose link the adviser adds to the slot.

Method

This section describes how the Consultation Scheduler will be analyzed, specified, designed, developed, tested and evaluated.

Requirement Analysis

The researchers observed the current process and interviewed its users. The current process is shown in Appendix B.

Requirements Documentation

The functional requirements are grouped by user in Table 1; the non-functional requirements follow in Table 2.

Table 1

Functional Requirements

ID	User	Requirement
FR-01	Adviser	Publish and edit weekly consultation slots
FR-02	Group	Book, cancel and reschedule a slot
FR-03	System	Email reminders one day and one hour before a meeting
FR-04	Adviser	Record topics and next steps after each consultation
FR-05	Group	View and export the consultation history
FR-06	Administrator	Manage adviser and group accounts

Table 2

Non-Functional Requirements

Characteristic	Requirement
Performance	A booking is confirmed on every open screen within two seconds
Usability	A group books a slot in under one minute on its first attempt
Security	Only university accounts can sign in; each group sees only its own records
Reliability	Two groups can never book the same slot

Design of Software, System, Product and Process

The system uses a client–server architecture: a React client, Firebase Authentication restricted to university accounts, Cloud Functions for every write, Firestore for data and a mail service for notifications. The main entities are ADVISER, GROUP, SLOT, BOOKING, CONSULTATION_NOTE.

Development and Testing

Development follows Scrum with two-week sprints. Testing proceeds from unit tests to integration tests on the Firebase Emulator Suite to acceptance testing with the functionality checklist (Appendix C).

Sample and Setting

Respondents are selected by purposive sampling as shown in Table 3.

Table 3

Respondents of the Study

Respondent group	Number	Instrument
Advisers	5	SUS, ISO/IEC 25010
Capstone students	30	SUS
IT experts	5	ISO/IEC 25010

Data Analysis Plan

SUS scores are computed per respondent and interpreted with the adjective scale. ISO/IEC 25010 ratings are summarized by weighted mean (4.21–5.00 Strongly Agree; 3.41–4.20 Agree; 2.61–3.40 Neutral; 1.81–2.60 Disagree; 1.00–1.80 Strongly Disagree).

Ethical Consideration

Participation is voluntary and respondents may withdraw at any time. Responses are anonymous, kept for the duration of the study and destroyed after the final defense, in accordance with the Data Privacy Act of 2012.

Results

Table 4

Functionality Checklist Results

Function	Result
Publish slots	Working
Book a slot	Working
Cancel and reschedule	Working
Email reminders	Working
Consultation notes	Working

Export history	Working
Account management	Working
No double booking	Working

Table 5

ISO/IEC 25010 Evaluation

Characteristic	Weighted mean	Interpretation
Functional Suitability	4.52	Strongly Agree
Performance Efficiency	4.31	Strongly Agree
Usability	4.44	Strongly Agree
Reliability	4.28	Strongly Agree

The mean SUS score was 81.5, which is above the benchmark of 68.

Double-booking test. Two groups booked the same slot 200 times in parallel; every attempt produced exactly one booking.

Discussion

Advisers rated functional suitability highest, citing that published slots ended the back-and-forth over meeting times. The lowest-rated item, reliability, reflected one test session in which reminders arrived late because of the mail provider's quota, which the group fixed by batching reminders.

Conclusion

The scheduler replaced message-based scheduling for the groups who used it, prevented double bookings, and was rated above average in usability (SUS 81.5).

Recommendations

1. Add calendar export (iCal) so advisers see bookings in their existing calendars.
2. Let advisers share one availability across all their groups.
3. Evaluate the system over a full semester instead of a four-week pilot.

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Appendix A

Cover Letter to Host/Locale

September 14, 2026

The Program Chair/Coordinator
School of Computing Faculty Room
Holy Angel University

Dear Sir/Madam:

We are students of the School of Computing working on our capstone project entitled "Thesis Adviser Consultation Scheduler." We respectfully ask permission to observe the current process at your office, interview your staff, and conduct a test session of the system. This letter is a placeholder for the dummy manuscript.

Respectfully yours,

Kilo Student
Lima Student
Mike Student
Golf Student

Noted by:
Prof. Alpha, Adviser

Appendix B

Current Process Flowchart

Request is made by message or on paper

Staff checks availability manually

Request is confirmed or declined

Activity takes place

Record is kept on paper, if at all

Appendix C

Functionality Checklist

Function	Working	Not working	Remarks
Publish slots			
Book a slot			
Cancel and reschedule			
Email reminders			
Consultation notes			
Export history			
Account management			
No double booking			

Appendix D

System Usability Scale Questionnaire

1 – Strongly Disagree, 2 – Disagree, 3 – Neutral, 4 – Agree, 5 – Strongly Agree

Statement	1	2	3	4	5
1. I think that I would like to use this system frequently.					
2. I found the system unnecessarily complex.					
3. I thought the system was easy to use.					
4. I think that I would need the support of a technical person to be able to use this system.					
5. I found the various functions in this system were well integrated.					
6. I thought there was too much inconsistency in this system.					
7. I would imagine that most people would learn to use this system very quickly.					
8. I found the system very cumbersome to use.					
9. I felt very confident using the system.					
10. I needed to learn a lot of things before I could get going with this system.					

Appendix E

ISO/IEC 25010 Software Quality Evaluation Form

Criteria	5	4	3	2	1
Functional Suitability					
The system meets the functional suitability requirements.					
Performance Efficiency					
The system meets the performance efficiency requirements.					
Usability					
The system meets the usability requirements.					
Reliability					
The system meets the reliability requirements.					